

Note :- All questions are compulsory.

Q. 1 **Attempt Any Two from the following.**

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- 1) Discuss object oriented approach.
- 2) Write a note on data abstraction.
- 3) Explain the concept of polymorphism with proper example.
- 4) Explain different forms of Inheritance.

Q. 2 **Attempt Any Two from the following.**

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- 1) Write a C++ program to show the use of pointers to object.
- 2) Explain how class members can be accessed?
- 3) How memory can be allocated for objects?
- 4) What are the parameterised constructor? Explain with example.

Q. 3 **Attempt Any Two from the following.**

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- 1) Explain the concept of binary operator overloading using friend function.
- 2) Write a C++ program to overload + operator.
- 3) How we can overload conversion operator? Explain with example.
- 4) Write a C++ program to overload relational operator.

Q. 4 **Attempt Any Two from the following.**

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- 1) Write a note on abstract class.
- 2) Explain how polymorphism can be achieved at compile time and run time?
- 3) What is virtual base class? When do we make a class virtual?
- 4) Describe various classes available for file operation.

Q. 5 **Attempt Any Two from the following.**

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- 1) Write a C++ program to show the use of assignment operator.
- 2) Write a note on stream manipulators.
- 3) Write a note on constructor and destructor.
- 4) Explain copy constructor.

Q. 6 **Attempt Any Two from the following.**

- 1) Write a note on iterator.
- 2) Write a C++ program to implement bubble sort algorithm using template function.
- 3) Write a note on standard C++ library.
- 4) List the three types of container.

Q. 7 **Attempt Any Three from the following:**

- 1) Write C++ program to implement bubble sort algorithm using template function.
- 2) Write C++ program to show the use of comparison operator.
- 3) Explain write () and read () function.
- 4) How we can overload subscript operator?
- 5) Write a note on Access specifier.
- 6) Write a note on class and object.

— The End —